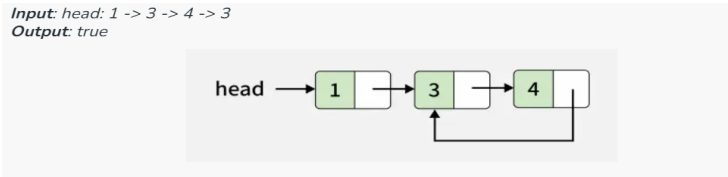
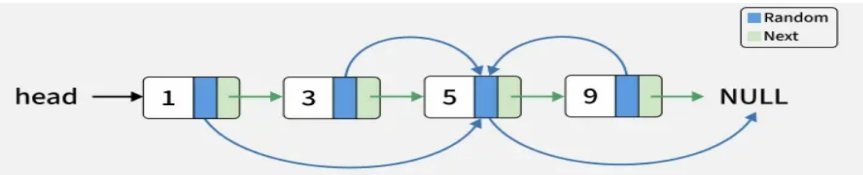
LAB 4 – Advance Linked List Operations

1. **Find the middle element of a singly linked list.**
   Example: 11 → 22 → 33 → 44 → 55 → NULL → Middle = 33

2. **Detect a loop in a singly linked list.**



3. **Remove duplicates from a sorted singly linked list.**
   Example: 2 → 2 → 2 → 5 → 5 → 9 → NULL → 2 → 5 → 9 → NULL
4. **Clone a linked list with next and random pointers.**



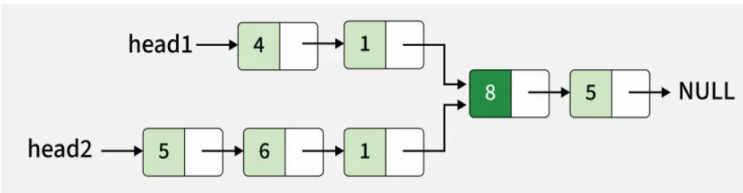
5. **Reverse a singly linked list in groups of size k.**
   Example: 1→2→3→4→5→6→7, k=3 → 3→2→1→6→5→4→7

6. **Check whether a singly linked list is a palindrome.**
   Example: 9 → 8 → 8 → 9 → NULL → Palindrome

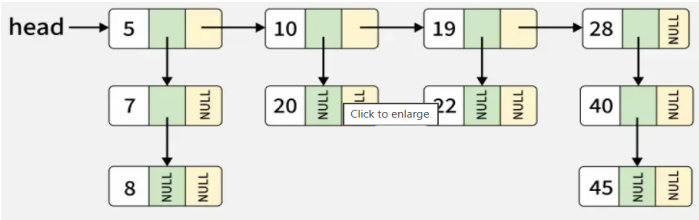
7. **Move the last node of a linked list to the front.**
   Example: 10 → 20 → 30 → 40 → NULL → 40 → 10 → 20 → 30 → NULL

8. **Merge two sorted singly linked lists into one sorted list.**
   Example: 1→4→8 and 2→3→9 → 1→2→3→4→8→9

9. **Find the intersection point of two singly linked lists.**



10. **Flatten a linked list having next and bottom pointers into a single sorted list.**



Output: *5 -> 7 -> 8 -> 10 -> 19 -> 20 -> 22 -> 28 -> 40 -> 45*

11. **Merge K sorted linked lists into one sorted linked list.**
    Example: L1=1→5, L2=2→6, L3=3→4 → 1→2→3→4→5→6